

MATH 231 – Linear Algebra I

Midterm 1 Practice #2 ■ Homework 1 and Homework 2 material

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About this practice exam

Every question here is a close relative of a problem from Homework 1 or Homework 2, with new numbers. The table on the last page lists the homework problems each question is modeled on, so you know where to look when one gives you trouble.

This is practice. It is not a statement about the format, the length, or the questions of the actual midterm.

Instructions

- **Parts A and B.** Circle one answer. No justification is needed; the space below each question is for scratch work.
- **Parts C, D and E.** Show your work in the space below each question. A correct answer with no derivation earns partial credit at best. If you run out of room, continue on the back of the page and say so.
- Leave answers in exact form: $\sqrt{13}$, $\pi/3$, $\ln 2$, $-\frac{4}{5} + \frac{7}{5}i$. No question needs a calculator.
- **Proofs.** Write down what you are assuming, justify each equality by naming the fact that licenses it, and end with a sentence saying what you have shown. A chain of equalities with no words is not a proof.

Part	Type	Questions
A	True or false	A.1–A.10
B	Multiple choice	B.1–B.8
C	Computation	C.1–C.6
D	Applications	D.1–D.4
E	Short proofs	E.1–E.5

Notation used throughout

Bold letters ($\mathbf{u}$, $\mathbf{v}$, $\mathbf{x}$) are vectors; plain letters (α , k , t) are scalars, that is, real numbers. For $\mathbf{u} = [u_1, \dots, u_n]$ and $\mathbf{v} = [v_1, \dots, v_n]$ in $\mathbb{R}^n$:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i, \quad \|\mathbf{v}\| = \|\mathbf{v}\|_2 = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}, \quad \|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i|, \quad \|\mathbf{v}\|_\infty = \max_{1 \leq i \leq n} |v_i|,$$

and each norm gives a distance $d_p(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p$.

For nonzero $\mathbf{u}, \mathbf{v}$, the **cosine similarity** is $\cos(\mathbf{u}, \mathbf{v}) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}$, and the angle $\theta \in [0, \pi]$ between $\mathbf{u}$ and $\mathbf{v}$ is the one with $\cos \theta = \cos(\mathbf{u}, \mathbf{v})$. The **unit vector** in the direction of $\mathbf{u}$ is $\hat{\mathbf{u}} = \mathbf{u}/\|\mathbf{u}\|$. For $\mathbf{v} \neq \mathbf{0}$, the **component** of $\mathbf{u}$ along $\mathbf{v}$ (a scalar) and the **projection** of $\mathbf{u}$ onto $\mathbf{v}$ (a vector) are

$$\text{comp}_{\mathbf{v}}(\mathbf{u}) = \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\|\mathbf{v}\|}, \quad \text{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\|\mathbf{v}\|^2} \mathbf{v},$$

and $\mathbf{u} - \text{proj}_{\mathbf{v}}(\mathbf{u})$ is the **orthogonal piece**.

In $\mathbb{R}^n$, $\mathbf{e}_i$ is the standard basis vector with a 1 in slot i and 0 elsewhere, and $\mathbf{1} = [1, 1, \dots, 1]$ is the all-ones vector.

A **subspace of $\mathbb{R}^2$** is a subset of $\mathbb{R}^2$ that is non-empty, closed under addition, and closed under scalar multiplication.

For a complex number $z = a + bi$ with $a, b \in \mathbb{R}$: $\text{Re}(z) = a$, $\text{Im}(z) = b$, $\bar{z} = a - bi$, $|z| = \sqrt{a^2 + b^2}$, and Euler's formula is $e^{i\theta} = \cos \theta + i \sin \theta$.

For a function $f(x, y)$, the **gradient** is $\nabla f = [\partial f / \partial x, \partial f / \partial y]$.

Part A. True or false

Circle **T** if the statement is always true, and **F** otherwise.

A.1 For all vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$, $\|\mathbf{u} - \mathbf{v}\| = \|\mathbf{v} - \mathbf{u}\|$. **T** **F**

A.2 For all complex numbers z and w , $|z + w| = |z| + |w|$. **T** **F**

A.3 $i^{99} = -i$. **T** **F**

A.4 If $\mathbf{u} \in \mathbb{R}^2$ is nonzero and α is a nonzero scalar, then $\text{proj}_{\alpha\mathbf{u}}(\mathbf{v}) = \text{proj}_{\mathbf{u}}(\mathbf{v})$ for every $\mathbf{v} \in \mathbb{R}^2$. **T** **F**

A.5 For all nonzero $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$, $\cos(-\mathbf{u}, \mathbf{v}) = \cos(\mathbf{u}, \mathbf{v})$. **T** **F**

A.6 For every $\mathbf{v} \in \mathbb{R}^2$, $\|\mathbf{v}\|_1 \leq 2\|\mathbf{v}\|_\infty$. **T** **F**

A.7 The set $\{(x, y) : x \geq 0 \text{ and } y \geq 0\}$ is a subspace of $\mathbb{R}^2$. **T** **F**

A.8 If $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ are linearly independent, then $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ are linearly independent. **T** **F**

A.9 If $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ are nonzero and $\langle \mathbf{u}, \mathbf{v} \rangle \neq 0$, then $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent. **T** **F**

A.10 If $f(x, y) = x e^y$, then $\nabla f(1, 0) = [1, 0]$. **T** **F**

Part B. Multiple choice

Circle the one correct answer.

B.1 Which vector has length 15 and points in the same direction as $[-3, 4]$?

- (A) $[-45, 60]$ (B) $[9, -12]$ (C) $[-9, 12]$ (D) $[-\frac{3}{5}, \frac{4}{5}]$

B.2 The quotient $\frac{4+3i}{2-i}$ equals

- (A) $2+i$ (B) $1+2i$ (C) $1-2i$ (D) $5+10i$

B.3 The set of complex numbers z with $|z+2i| \leq 1$ is

- (A) the circle of radius 1 centred at $-2i$
(B) the closed disc of radius 1 centred at $2i$
(C) the closed disc of radius 1 centred at $-2i$
(D) the open disc of radius 1 centred at $-2i$

B.4 The angle between $[2, 0]$ and $[-1, \sqrt{3}]$ is

(A) $\pi/6$

(B) $\pi/3$

(C) $\pi/2$

(D) $2\pi/3$

B.5 For $\mathbf{u} = [2, 3]$ and $\mathbf{v} = [3, 4]$, the component $\text{comp}_{\mathbf{v}}(\mathbf{u})$ is

(A) $\frac{18}{5}$

(B) $\frac{18}{25}$

(C) $[\frac{54}{25}, \frac{72}{25}]$

(D) $\frac{18}{\sqrt{13}}$

B.6 Which pair is a basis of $\mathbb{R}^2$?

(A) $[2, -4]$ and $[-1, 2]$

(B) $[3, 6]$ and $[1, 2]$

(C) $[0, 0]$ and $[1, 3]$

(D) $[1, 2]$ and $[2, 1]$

B.7 In $\mathbb{R}^9$, the norms $\|\mathbf{1}\|_1$, $\|\mathbf{1}\|_2$ and $\|\mathbf{1}\|_\infty$ are, in that order,

(A) 1, 3, 9

(B) 9, 3, 1

(C) 9, 81, 1

(D) 9, 9, 9

B.8 For $f(x, y) = (x - 3y)^4$, the partial derivative $\partial f / \partial y$ is

(A) $-12(x - 3y)^3$

(B) $4(x - 3y)^3$

(C) $-3(x - 3y)^3$

(D) $12(x - 3y)^3$

Part C. Computation

Show your work. Leave every answer in exact form.

Problem C.1 (Vectors in the plane).

Let $\mathbf{u} = [-1, 3]$ and $\mathbf{v} = [4, 1]$ be vectors in $\mathbb{R}^2$, and let $\alpha = -3$ be a scalar.

(a) Compute $\mathbf{u} - \mathbf{v}$, $3\mathbf{u} + 2\mathbf{v}$, and $\alpha\mathbf{u}$, showing the componentwise step.

(b) Compute $\|\mathbf{u}\|$, $\|\mathbf{v}\|$, $\|\mathbf{u} + \mathbf{v}\|$ and $\|\alpha\mathbf{u}\|$.

(c) For this pair, does the triangle inequality $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ hold strictly ($<$) or with equality ($=$)?

Problem C.1, continued

- (d) Find every scalar β for which $\|\mathbf{u} + \beta\mathbf{u}\| = \|\mathbf{u}\| + \|\beta\mathbf{u}\|$. You may use $\|\gamma\mathbf{x}\| = |\gamma| \|\mathbf{x}\|$ for every scalar γ and every vector $\mathbf{x}$.

- (e) The arrow from $(2, -1)$ to $(1, 2)$ and the arrow from $(0, 4)$ to $(3, 3)$ both have length $\sqrt{10}$. Write each arrow as a vector, using *head minus tail*, and decide which of them, if either, is the vector $\mathbf{u}$. Justify your answer with the definition of equality of vectors.

Problem C.2 (Complex arithmetic).

Let $z = 4 - i$ and $w = 2 + 3i$. Give every answer that is not a modulus in the form $a + bi$ with a and b real.

(a) Compute $z + w$, $z - w$, and zw .

(b) Compute $\overline{w}$, $w\overline{w}$, and $\frac{z}{w}$.

(c) Compute $|z|$, $|w|$ and $|zw|$, and compare $|zw|$ with $|z| |w|$.

Problem C.2, continued

(d) Compute $(1 - i)^2$ and $(1 - i)^6$.

(e) Use Euler's formula to write $e^{3\pi i/4}$ and $e^{-i\pi/3}$ in the form $a + bi$.

(f) The point $(-3, 4)$ is rotated 90° *clockwise* about the origin. Find the coordinates of the rotated point by a single multiplication of complex numbers.

Problem C.3 (Dot products, angles, and orthogonality).

For each pair below, compute $\langle \mathbf{u}, \mathbf{v} \rangle$, $\|\mathbf{u}\|$, $\|\mathbf{v}\|$ and $\cos(\mathbf{u}, \mathbf{v})$. Then give the angle θ between the two vectors, and state which of *parallel*, *antiparallel*, *orthogonal* or *none of the three* describes the pair.

	$\mathbf{u}$	$\mathbf{v}$		$\mathbf{u}$	$\mathbf{v}$
(a)	$[3, -2]$	$[4, 6]$	(c)	$[2, 1]$	$[-3, 1]$
(b)	$[-1, 2]$	$[-3, 6]$	(d)	$[\sqrt{3}, 1]$	$[0, 2]$

Problem C.3, continued

- (e) Find every real number k for which $[k, 2]$ is orthogonal to $[k + 1, -1]$.
- (f) Find every real number k for which the angle between $[1, k]$ and $[0, 1]$ is $\pi/6$. You will want to square both sides; check every candidate in the original equation, and discard any that fail it.
- (g) In $\mathbb{R}^4$, let $\mathbf{u} = [1, -1, 2, 0]$ and $\mathbf{w} = [1, 0, 1, 1]$. Compute $\langle \mathbf{u}, \mathbf{w} \rangle$, $\|\mathbf{u}\|$, $\|\mathbf{w}\|$, and the angle between $\mathbf{u}$ and $\mathbf{w}$.

Problem C.4 (Bases, independence, and coordinates).

Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ are **linearly independent** when $\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{0}$ forces $\alpha = \beta = 0$, and **linearly dependent** otherwise. For $\mathbf{u} = [u_1, u_2]$ and $\mathbf{v} = [v_1, v_2]$, put $D = u_1v_2 - u_2v_1$. The pair is linearly independent, and so a basis of $\mathbb{R}^2$, exactly when $D \neq 0$ (Vector Unit, Part 4, §18.5).

- (a) Let $\mathbf{u} = [2, 1]$ and $\mathbf{v} = [-1, 3]$. Compute D . Then find the coordinates α, β of $\mathbf{w} = [3, 5]$ and of $\mathbf{w} = [4, -5]$ against the basis $\{\mathbf{u}, \mathbf{v}\}$ of $\mathbb{R}^2$, by setting up the linear system $\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{w}$ and eliminating by hand. Check each answer by substituting back.

- (b) Show that $[-6, 9]$ and $[4, -6]$ are linearly dependent by giving scalars α, β , not both zero, with $\alpha[-6, 9] + \beta[4, -6] = \mathbf{0}$. Describe their span as a set $\{(x, y) : \dots\}$, give a basis of it, and state its dimension.

Problem C.4, continued

- (c) Find every real number k for which $[2, k]$ and $[k, 8]$ are linearly dependent. For each such k , give the equation of the line through the origin that the two vectors span.

- (d) Let $S = \{[1, 2], [3, -1], [4, 1]\}$. Write $[5, 3]$ as a linear combination of the three vectors of S in two different ways. A basis of $\mathbb{R}^2$ must span $\mathbb{R}^2$ and be linearly independent; S spans $\mathbb{R}^2$. Use your two combinations to show which of the two conditions S fails.

Problem C.5 (Orthogonal bases).

If $\mathbf{b}_1, \dots, \mathbf{b}_k$ are nonzero, mutually orthogonal, and form a basis, then the coefficient of $\mathbf{b}_j$ in a vector $\mathbf{w}$ is $\langle \mathbf{b}_j, \mathbf{w} \rangle / \|\mathbf{b}_j\|^2$ (Vector Unit, Part 5, §21.6).

- (a) Let $\mathbf{u} = [3, -1]$ and $\mathbf{v} = [1, 3]$. Check that they are nonzero and orthogonal. Find the coordinates of $[-1, 7]$ against the basis $\{\mathbf{u}, \mathbf{v}\}$ of $\mathbb{R}^2$ by projection, and check them by substituting back.

- (b) In $\mathbb{R}^3$, let $\mathbf{u} = [1, 1, 0]$, $\mathbf{v} = [1, -1, 2]$ and $\mathbf{w} = [-1, 1, 1]$. Check that they are mutually orthogonal. Find, by projection, the coefficients α, β, γ with $\mathbf{x} = \alpha\mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w}$ for $\mathbf{x} = [4, 0, 1]$, and substitute back to confirm. Then verify that $\|\mathbf{x}\|^2 = \alpha^2\|\mathbf{u}\|^2 + \beta^2\|\mathbf{v}\|^2 + \gamma^2\|\mathbf{w}\|^2$.

Problem C.5, continued

- (c) Let $\mathbf{u} = [1, 1]$ and $\mathbf{v} = [3, 1]$. Show that they are not orthogonal. Compute $\mathbf{v}' = \mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v})$ and check that $\mathbf{v}'$ is orthogonal to $\mathbf{u}$. Find the coordinates of $[5, 1]$ against the orthogonal basis $\{\mathbf{u}, \mathbf{v}'\}$ by projection. Then rewrite the result as a combination of the original $\mathbf{u}$ and $\mathbf{v}$, and check it.

Problem C.6 (Derivatives, partial derivatives, and the gradient).

(a) Differentiate each function, and name the rule or rules you use.

$$(i) \ x^3 \ln x \quad (ii) \ e^{-x} \cos x \quad (iii) \ \ln(1 + e^{2x}) \quad (iv) \ \frac{x}{1 + x^2}$$

(b) Solve $e^{2x} + e^x - 6 = 0$.

Problem C.6, continued

- (c) Solve $\ln(x+1) + \ln(x-2) = \ln 4$. Reject any candidate that does not solve the original equation, and say why it fails.

- (d) Compute $\partial f/\partial x$ and $\partial f/\partial y$ for

$$f(x, y) = 2x^4y - 3xy^3 + 5 \quad \text{and} \quad f(x, y) = x e^{-xy}.$$

- (e) Let $h(x, y) = x^2y - 2xy + y^2$. Compute ∇h , the vector $\nabla h(0, 2)$ and its norm, and find every point (x, y) at which $\nabla h = \mathbf{0}$.

Part D. Applications

Show your work, and give units wherever the problem has them.

Problem D.1 (Work, and a box on a ramp).

A force is a vector $\mathbf{F} \in \mathbb{R}^2$: its direction is the way it pushes, and its norm $\|\mathbf{F}\|$ is how hard. When a constant force $\mathbf{F}$ acts on an object that moves in a straight line with displacement $\mathbf{s} \in \mathbb{R}^2$, the **work** done by the force is the scalar $W = \langle \mathbf{F}, \mathbf{s} \rangle$, in joules when $\mathbf{F}$ is in newtons and $\mathbf{s}$ in metres. So $\mathbf{F}$, $\mathbf{s}$, $\mathbf{d}$, $\mathbf{G}$ and $\mathbf{P}$ below are vectors, while W , every norm, and every component are scalars.

A box sits on a ramp that rises in the direction $\mathbf{d} = [3, 1]$, and gravity pulls on it with the constant force $\mathbf{G} = [0, -20]$.

- (a) Compute $\text{comp}_{\mathbf{d}}(\mathbf{G})$, $\text{proj}_{\mathbf{d}}(\mathbf{G})$, and the orthogonal piece $\mathbf{G} - \text{proj}_{\mathbf{d}}(\mathbf{G})$. Check that the orthogonal piece is orthogonal to $\mathbf{d}$.

- (b) Compute the norms of the two pieces. One pulls the box down the ramp and the other presses it into the ramp's surface: which is which? Check that $\|\mathbf{G}\|^2$ is the sum of the squares of the two norms.

Problem D.1, continued

- (c) The box is pushed up the ramp from $(0, 0)$ to $(6, 2)$. Compute the work done by gravity, and say what its sign means.

- (d) Over the same displacement, a worker pushes horizontally with the force $\mathbf{P} = [25, 0]$. Compute the work the worker does in two ways: as $\langle \mathbf{P}, \mathbf{s} \rangle$, and as $\text{comp}_{\mathbf{s}}(\mathbf{P}) \cdot \|\mathbf{s}\|$. Then give a nonzero force that does no work at all over this displacement, and justify it with $\langle \mathbf{F}, \mathbf{s} \rangle = \|\mathbf{F}\| \|\mathbf{s}\| \cos \theta$.

Problem D.2 (Which customer is nearest? It depends on the metric).

A depot sits at $P = (1, -2)$, and three customers at $A = (1, 5)$, $B = (-4, -7)$ and $C = (3, -8)$.

(a) Make a table of the distances d_1 , d_2 and d_∞ from P to each customer.

(b) Three vehicles leave the depot: a *taxi* that must follow a rectangular street grid, a *drone* that flies in a straight line, and an *overhead crane* whose east–west and north–south motors run at the same time and at the same speed. Which of d_1 , d_2 , d_∞ measures the trip for each vehicle, and which customer is nearest for each?

(c) On one set of axes, sketch and label the three sets $\{\mathbf{v} \in \mathbb{R}^2 : \|\mathbf{v}\|_1 = 2\}$, $\{\mathbf{v} \in \mathbb{R}^2 : \|\mathbf{v}\|_2 = 2\}$ and $\{\mathbf{v} \in \mathbb{R}^2 : \|\mathbf{v}\|_\infty = 2\}$. For each one, find the point where it crosses the ray $y = x$, $x > 0$.

Problem D.3 (Comparing documents).

One way to compare documents is to count words. Fix the vocabulary *goal*, *team*, *coach*, *vote*, *budget*, and record each document as the vector in $\mathbb{R}^5$ of its word counts, in that order:

$$\mathbf{a} = [1, 2, 2, 0, 0], \quad \mathbf{b} = [3, 6, 6, 0, 0], \quad \mathbf{c} = [0, 0, 0, 4, 3], \quad \mathbf{d} = [2, 1, 0, 2, 0].$$

(a) Compute $\cos(\mathbf{a}, \mathbf{b})$, $\cos(\mathbf{a}, \mathbf{c})$ and $\cos(\mathbf{a}, \mathbf{d})$.

(b) Compute the Euclidean distances $d_2(\mathbf{a}, \mathbf{b})$, $d_2(\mathbf{a}, \mathbf{c})$ and $d_2(\mathbf{a}, \mathbf{d})$.

(c) Rank $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ from closest to farthest from $\mathbf{a}$, once by cosine similarity and once by Euclidean distance. Document $\mathbf{b}$ is document $\mathbf{a}$ pasted in three times, and $\mathbf{c}$ shares no words with $\mathbf{a}$. Which of the two measures gives the sensible ranking for comparing what documents are about, and why?

Problem D.4 (How kinetic energy responds to mass and speed).

An object of mass m kilograms moving at speed v metres per second has kinetic energy $K(m, v) = \frac{1}{2}mv^2$ joules.

(a) Compute $\partial K/\partial m$ and $\partial K/\partial v$.

(b) Evaluate both partial derivatives for a car with $m = 1000$ and $v = 20$, and give their units.

(c) Use (b) to estimate the change in kinetic energy when a 50 kg passenger gets in with the speed held fixed, and when the speed rises by 1 m/s with the mass held fixed. Which change adds more energy?

Part E. Short proofs

None of these needs a clever idea. Write down what you are assuming, justify each step by naming the law or fact that licenses it, and end with a sentence saying what you have shown.

The laws of the dot product, from §11.8 of Vector Unit, Part 2 (*Orientation, the Dot Product, and Projection*), hold for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ and all scalars α :

- | | | |
|-----------|---|-----------------------|
| D1 | $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ | symmetry |
| D2 | $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$ | additivity |
| D3 | $\langle \mathbf{u}, \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle$ | homogeneity |
| D4 | $\langle \mathbf{u}, \mathbf{u} \rangle = \ \mathbf{u}\ ^2 \geq 0$, with equality only for $\mathbf{u} = \mathbf{0}$ | positive definiteness |

Problem E.1 (A law of vector addition, in components).

Let $\mathbf{u} = [u_1, u_2]$, $\mathbf{v} = [v_1, v_2]$ and $\mathbf{w} = [w_1, w_2]$ be vectors in $\mathbb{R}^2$. Working in components, prove the law **A2** of §8.4 of Vector Unit, Part 1:

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w}).$$

Name the property of the real numbers that the proof rests on.

Problem E.2 (The real part, and a dot product in disguise).

Let $z = a + bi$ and $w = c + di$ with $a, b, c, d \in \mathbb{R}$.

(a) Prove that $z + \bar{z} = 2 \operatorname{Re}(z)$ and $z - \bar{z} = 2i \operatorname{Im}(z)$.

(b) Prove that $z\bar{w} + \bar{z}w = 2(ac + bd)$. So $z\bar{w} + \bar{z}w$ is always a real number, and it is twice the dot product of the vectors $[a, b]$ and $[c, d]$.

(c) Check (b) for $z = 1 + 2i$ and $w = 3 - i$ by computing $z\bar{w}$ and $\bar{z}w$ separately and adding them.

Problem E.3 (The orthogonal piece, and an exact accounting).

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ with $\mathbf{v} \neq \mathbf{0}$, and put $c = \langle \mathbf{v}, \mathbf{u} \rangle / \|\mathbf{v}\|^2$, so that $\text{proj}_{\mathbf{v}}(\mathbf{u}) = c \mathbf{v}$.

(a) Using only **D1–D4** — no components — prove that $\langle \mathbf{v}, \mathbf{u} - c\mathbf{v} \rangle = 0$.

Hint. Write $\mathbf{u} - c\mathbf{v} = \mathbf{u} + (-c)\mathbf{v}$ and expand with **D2** and **D3**.

(b) The **Pythagorean identity** (Vector Unit, Part 5, §23) says that if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$, then $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$. Use it and (a) to prove that

$$\|\mathbf{u}\|^2 = \|\text{proj}_{\mathbf{v}}(\mathbf{u})\|^2 + \|\mathbf{u} - \text{proj}_{\mathbf{v}}(\mathbf{u})\|^2.$$

Hint. Part (a) is about $\mathbf{v}$, but the identity needs $\langle c\mathbf{v}, \mathbf{u} - c\mathbf{v} \rangle = 0$. Move the c out with **D1** and **D3**.

Problem E.4 (A weighted norm).

For $\mathbf{v} = [v_1, v_2] \in \mathbb{R}^2$, define

$$\|\mathbf{v}\|_w = |v_1| + 2|v_2|.$$

A function on $\mathbb{R}^2$ is a **norm** when it satisfies the three properties of §15.1 of Vector Unit, Part 3, for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ and all scalars α :

N1 $\|\mathbf{v}\| \geq 0$, and $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$ positive definiteness

N2 $\|\alpha\mathbf{v}\| = |\alpha| \|\mathbf{v}\|$ absolute homogeneity

N3 $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ the triangle inequality

Prove that $\|\cdot\|_w$ satisfies **N1**, **N2** and **N3**. You may use two facts about real numbers a and b : $|ab| = |a| |b|$, and $|a + b| \leq |a| + |b|$.

Hint. For **N1**, a sum of two non-negative numbers is zero only when both of them are.

Problem E.5 (A span is a subspace).

Fix two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$, and let

$$\text{span}\{\mathbf{u}, \mathbf{v}\} = \{ \alpha \mathbf{u} + \beta \mathbf{v} : \alpha, \beta \in \mathbb{R} \}.$$

- (a) Prove that $\text{span}\{\mathbf{u}, \mathbf{v}\}$ is a subspace of $\mathbb{R}^2$: show that it is non-empty, closed under addition, and closed under scalar multiplication. You may use the fact that $0\mathbf{x} = \mathbf{0}$ for every vector $\mathbf{x}$, and these laws of §8.4 of Vector Unit, Part 1, for all vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$ and scalars γ, δ :

$$\mathbf{A1} \quad \mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$$

$$\mathbf{A2} \quad (\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$$

$$\mathbf{A3} \quad \mathbf{x} + \mathbf{0} = \mathbf{x}$$

$$\mathbf{S1} \quad \gamma(\mathbf{x} + \mathbf{y}) = \gamma\mathbf{x} + \gamma\mathbf{y}$$

$$\mathbf{S2} \quad (\gamma + \delta)\mathbf{x} = \gamma\mathbf{x} + \delta\mathbf{x}$$

$$\mathbf{S3} \quad \gamma(\delta\mathbf{x}) = (\gamma\delta)\mathbf{x}$$

- (b) Prove that if a vector $\mathbf{w} \in \mathbb{R}^2$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$, then $\mathbf{w}$ is orthogonal to every vector in $\text{span}\{\mathbf{u}, \mathbf{v}\}$. Use the laws **D2** and **D3**.

Where each question comes from

Each question is modeled on the homework problems listed beside it. If one gives you trouble, rework those problems first. HW1 is Homework 1 and HW2 is Homework 2.

Question	Homework problems	Question	Homework problems
A.1	HW1 B.1, B.3	C.1	HW1 A.1–A.4, B.3, B.5
A.2	HW1 C.5, D.3	C.2	HW1 C.2–C.7
A.3	HW1 C.3	C.3	HW1 E.1, E.3; HW2 A.2, E.1
A.4	HW2 A.6	C.4	HW2 C.1–C.3
A.5	HW1 E.3	C.5	HW2 D.1, D.2, D.4
A.6	HW2 B.1, E.7	C.6	HW2 F.1, F.2, F.4, F.6
A.7	HW2 C.4	D.1	HW1 E.4; HW2 A.3
A.8	HW2 C.5	D.2	HW2 B.1, B.2
A.9	HW2 C.2, D.2	D.3	HW2 E.4
A.10	HW2 F.4, F.6	D.4	HW2 F.5
B.1	HW2 A.1	E.1	HW1 B.2
B.2	HW1 C.2	E.2	HW1 C.4, D.1, D.2
B.3	HW1 C.5	E.3	HW1 E.7; HW2 D.4
B.4	HW1 E.3; HW2 A.2	E.4	HW2 B.3, E.2
B.5	HW1 E.4	E.5	HW2 C.4, C.5, D.5
B.6	HW2 C.1, C.2		
B.7	HW2 E.3		
B.8	HW2 F.4		

Questions about any problem here belong in the class Discord, where everyone can see the answer. Questions that are specific to your own work are better brought to office hours, Thursdays 3:00–4:00 PM in Kiely Hall 508. Please email ahead.